

1. INTRODUCTION

Brain-Computer Interface (BCI) technology enables direct communication between the human brain and external devices, allowing cognitive or mental states to control hardware without physical input. This emerging interdisciplinary domain integrates neuroscience, biomedical instrumentation, signal processing, embedded systems, and artificial intelligence. In this project, we propose a low-cost, do-it-yourself (DIY) EEG-based BCI system enhanced with an intelligent assistant named "Rocker," which interprets brain signals to trigger actions such as turning on appliances, playing music, and supporting relaxation or focus. This system leverages Arduino for signal acquisition and processing, a homemade amplifier circuit for signal enhancement, and a Python-based NLP engine to deliver intelligent responses based on mental states.

1.1 Background

Traditional interfaces such as touchscreens, keyboards, and voice inputs are limited by the user's physical capability and environmental conditions. Brain-Computer Interfaces offer an alternative pathway by capturing electrical activity from the brain (via EEG) and using it as a control signal. EEG is a non-invasive technique that measures voltage fluctuations resulting from ionic current flows within neurons. These signals, though weak, carry valuable information about cognitive states like attention, drowsiness, and motor imagery.

Over the past two decades, research in BCI has advanced significantly. However, commercial BCI systems remain expensive, and their integration with AI assistants is largely unexplored in DIY domains. This project is grounded in the vision of democratizing brain-tech by demonstrating how basic materials and open-source platforms can produce powerful outcomes. *The integration of a custom voice assistant ("Rocker") allows natural interaction and automation, bringing futuristic functionality to home and educational use cases.*

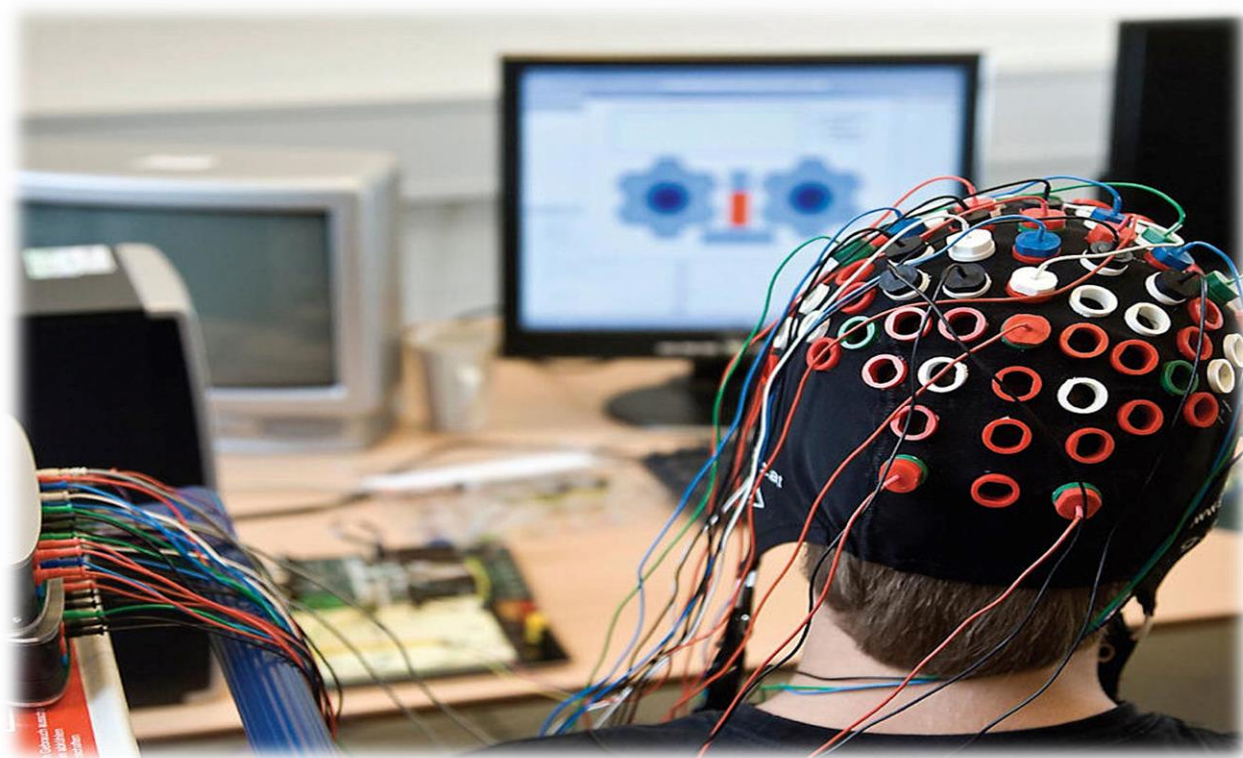


Fig 1.1 Brain Computer Interface, EEG

1.2 Problem Statement

Despite breakthroughs in neuroscience and AI, BCIs are not yet accessible to hobbyists or students due to the high cost and complexity of EEG hardware. The problem can be formulated as follows:

"Can we design a budget-friendly, safe, and intelligent brain-computer interface that reads mental states using household materials, processes them with Arduino, and triggers actions via an AI voice assistant?"

Challenges include:

- Capturing clean EEG signals with low-cost sensors.
- Filtering noise and amplifying weak signals.
- Interpreting blinks, relaxation, and focus states reliably.
- Mapping these states to meaningful outputs (e.g., LED, music, fan).
- Building an NLP-based assistant that understands commands initiated through brain activity.



Despite significant advances in **neuroscience**, **artificial intelligence (AI)**, and **wearable technologies**, **brain-computer interfaces (BCIs)** remain largely out of reach for students, hobbyists, and independent researchers. This is primarily due to the **prohibitive cost, hardware complexity, and steep learning curve** associated with commercial-grade EEG systems. These systems often rely on high-precision medical-grade sensors, proprietary software, and expensive signal processing equipment, limiting their accessibility to academic labs or professional institutions.

However, with the increasing availability of **affordable microcontrollers like Arduino**, open-source platforms, and the democratization of AI tools, a question arises:

"Can we design a budget-friendly, safe, and intelligent brain-computer interface that reads mental states using household materials, processes them with Arduino, and triggers actions via an AI voice assistant?"

This research and development initiative aims to explore this question by building a low-cost, accessible BCI system that bridges **mental activity and machine control**, using **EEG signal acquisition, signal conditioning, machine learning, and natural language processing (NLP)**.

Key Challenges and Objectives

1. **Capturing Clean EEG Signals with Low-Cost Sensors**

EEG signals are inherently **low-amplitude (10–100 μV)** and highly vulnerable to noise from muscle activity, eye movements, and electrical appliances. Using affordable materials (e.g., dry electrodes, conductive fabric, DIY headsets), the challenge is to capture meaningful brain signals such as alpha, beta, and gamma waves reliably.

2. **Filtering Noise and Amplifying Weak Signals**

The raw EEG signals must be **amplified and filtered** before further processing. This involves:

- Designing **low-noise differential amplifiers**.
- Implementing **bandpass filters** to isolate relevant brainwave frequencies.
- Suppressing artifacts through **hardware shielding** and **signal processing algorithms**.

These tasks must be achieved using cost-effective components like **LM358 op-amps**, passive RC networks, and compact PCBs.

3. **Interpreting Blinks, Relaxation, and Focus States Reliably**

A core objective is to decode brain patterns associated with intentional states:

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- **Blink detection** for triggering discrete commands.
 - **Relaxation detection** (associated with alpha waves) for controlling devices like lights or music.
 - **Focus detection** (linked to beta activity) for initiating tasks like fan control or data logging.
- The challenge lies in designing **lightweight machine learning algorithms** or threshold-based classifiers that can run on limited computational resources like the Arduino or edge devices.

4. Mapping Brain States to Meaningful Outputs

Once brain states are interpreted, they must be translated into **real-world actions** such as:

- Switching **LEDs** on or off.
 - Activating a **motor or fan**.
 - Playing or pausing **music** based on mental state transitions.
- This requires integrating Arduino with actuators and ensuring reliable communication between the signal processing pipeline and control logic.

5. Building an NLP-Based Assistant Triggered by Brain Activity

One of the novel goals of this project is to develop a **voice-controlled AI assistant** (e.g., using Python and NLP libraries like spaCy or transformers) that is activated through a **brain signal-based trigger** (such as a blink or a focus state). Once activated, the assistant can:

- Receive and interpret **verbal commands**.
- Provide **auditory feedback**.
- Control connected devices or fetch information.

The innovation lies in the seamless integration of **biological input (EEG)** with **artificial intelligence (NLP)** for an immersive and intuitive human-machine interaction.

Vision

The long-term vision is to democratize BCI technology by enabling **low-cost, DIY solutions** for students, developers, and enthusiasts. This could revolutionize assistive technologies, educational tools, and even creative applications such as **mind-controlled art, music synthesis, or gaming**—making brain-machine interfaces **accessible to all**.

1.3 Objectives of the Project

The primary goal of this project is to design and implement an **innovative, budget-friendly, and functional Brain-Computer Interface (BCI) system** that leverages **household materials, low-cost electronics, and AI technologies** to decode mental states and map them to meaningful real-world actions. This system is designed to be **accessible for students, hobbyists, and developers**, serving as both a practical learning tool and a potential prototype for future assistive technologies.

The **core objectives** of the project are outlined below:

1. Develop a DIY EEG Acquisition Setup

- Construct a **cost-effective EEG signal acquisition mechanism** using readily available materials such as **coin-shaped aluminum electrodes, conductive fabric, or foil**, strategically positioned on the scalp using a custom-designed headset.

- Interface these electrodes with the **Arduino Uno** via the **analog input pin (A0)**, enabling real-time analog signal sampling.
- Ensure safe electrical contact and reliable signal pickup using a minimal component footprint, all while maintaining user comfort and portability.

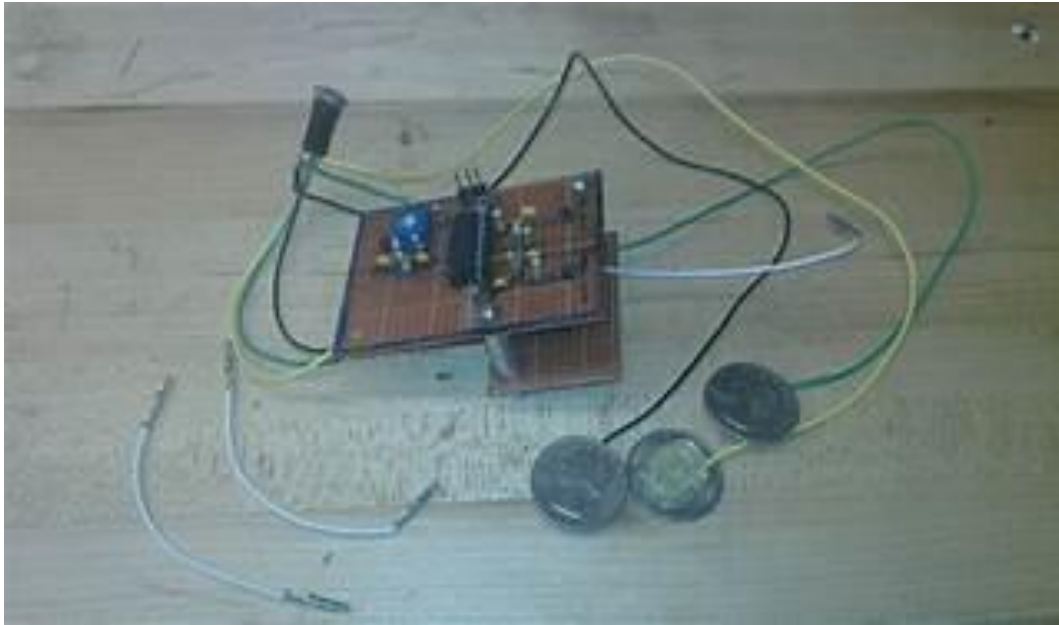


Fig 1.2 Easy EEG sensor setup

2. Design a Signal Amplification and Conditioning Circuit

- Build a **two-stage amplification circuit** using the **LM358 dual op-amp**, configured as a differential amplifier followed by an active bandpass filter.
- Optimize the amplification to elevate the microvolt-level EEG signals into a readable range (millivolts) without saturating or distorting the signal.
- Implement **noise filtering techniques** (such as snubber networks and RC filters) to isolate meaningful EEG bands (0.5–35 Hz), particularly targeting **alpha**, **beta**, and **gamma waves**, which correlate with relaxation, focus, and cognitive processing.

3. Implement Brain State Detection Algorithms

- Develop algorithms to identify **blinks**, **focus**, and **relaxation states** by analyzing:
 - **Voltage thresholds**
 - **Sudden signal peaks**
 - **Time-based pulse patterns**
- Employ lightweight statistical or threshold-based classifiers to enable **real-time detection** on microcontroller-level processing power.
- Create a flexible detection pipeline that is adaptive to different users and ambient noise conditions.

4. Interface Brain States with a Custom NLP Assistant ("Rocker")

- Integrate the EEG decoding system with a custom-built **AI voice assistant named "Rocker"**, designed using **Python, speech recognition, and NLP libraries**.
- Configure Rocker to respond to various **mental triggers** detected by the BCI system:
 - **Double blink** → Toggle an **LED light** (basic control)
 - **Rapid multiple blinks** → Start playing **rap music** (entertainment trigger)
 - **High mental focus** → Initiate a **productive environment** (e.g., turn on study lights or block distractions)
 - **Relaxed/calm state** → Activate a **fan** for comfort (environmental response)
- This integration illustrates the potential for **hands-free control of smart devices using thoughts and expressions**, making the interaction more intuitive and human-centric.

5. Demonstrate a Real-World Use-Case of Brain-Controlled Automation

- Construct a prototype system that translates **cognitive and emotional states** into **practical actions**, creating an immersive demonstration of **brain-controlled home automation**.
- Use case scenarios include:
 - Controlling household appliances (lights, fans, music) without physical movement
 - Enhancing accessibility for differently-abled individuals
 - Exploring new modes of human-computer interaction (HCI) for gaming, art, and productivity.

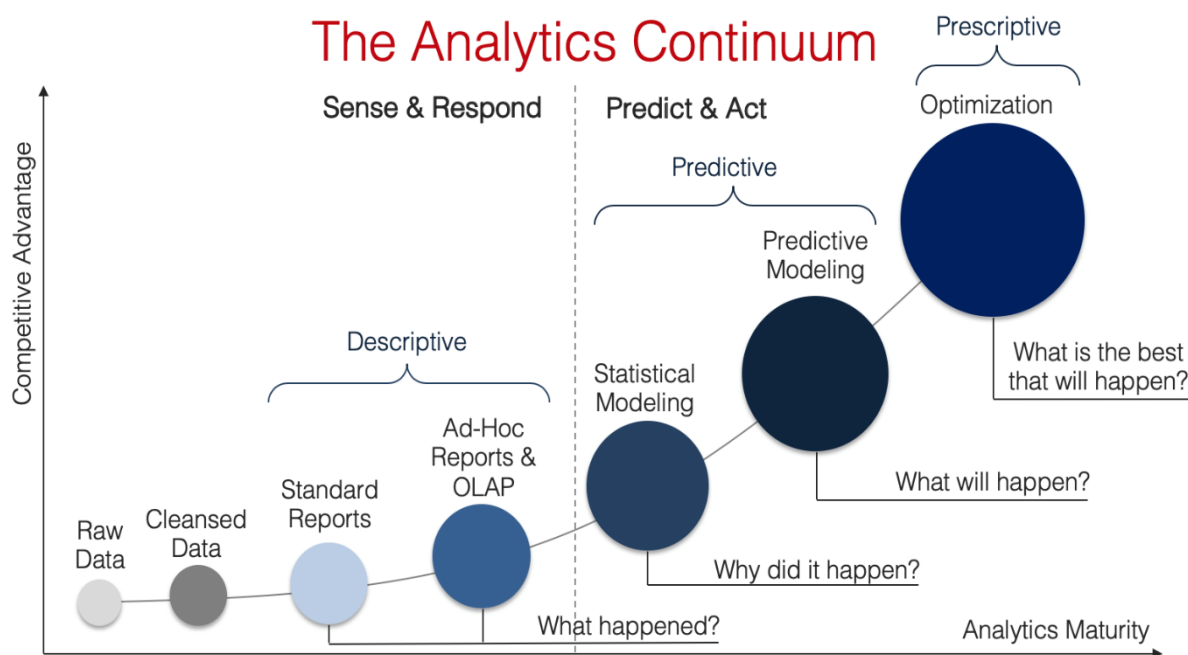


Fig 1.3 Analytics Trend

6. Explore the Potential of Combining AI with BCI

- Investigate how **Artificial Intelligence**, particularly **Natural Language Processing (NLP)** and **machine learning**, can augment brain-computer interfaces to become more **context-aware, adaptable, and intelligent**.
- Promote the use of AI-driven BCI systems for:
 - **Accessibility** – empowering users with physical impairments
 - **Productivity** – enabling distraction-free, hands-free digital environments
 - **Inclusion** – offering alternative input mechanisms for communication and interaction
- Contribute to the growing field of **neurotechnology democratization**, proving that impactful innovation can emerge even with constrained budgets and resources.

1.4 Motivation

The motivation behind this project stems from the need for inclusive technology and the desire to push the boundaries of DIY innovation.

- **Personal drive:** A passion for embedded systems, neuroscience, and AI inspired the vision of developing a thought-controlled assistant.
- **Assistive technology:** Physically challenged individuals often find traditional interfaces difficult to use. A low-cost, BCI-driven control system could empower them to interact with the digital world effortlessly.
- **Innovation and Learning:** Building a brain-controlled AI assistant offers a rare opportunity to combine signal processing, hardware design, and machine intelligence in one holistic project.
- **Educational outreach:** Demonstrating that advanced technology can be replicated affordably with accessible materials encourages innovation at the grassroots level.

1.5 Scope of Work

The project scope includes the following:

1. Hardware Development

This stage involves creating the physical components required to capture brain signals (EEG), amplify them, and pass them to a microcontroller for further processing.

• Building Dry Electrodes using Coins/Aluminum:

- Instead of using expensive clinical-grade electrodes, **low-cost dry electrodes** were fabricated using **metallic coins or aluminum foil**.
- These materials can effectively conduct electrical signals from the scalp.
- The electrodes are connected via conductive wires and mounted on a headband or cap for stable placement.
- Dry electrodes don't require gel or skin preparation, making the setup more user-friendly.

• Assembling a Signal Amplification Circuit using LM358 or Similar Op-Amp:

- Brain signals (EEG) are extremely **weak**, typically in the range of **microvolts (μV)**.
- An **operational amplifier (op-amp)** like **LM358** is used to amplify these signals to a level that can be read by the Arduino.
- The circuit often includes:
 - **Gain resistors** to control amplification factor (e.g., 100x–1000x),
 - **Filtering components** (capacitors/resistors) to remove unwanted noise.
- The output of this amplifier is a voltage signal proportional to the brainwave activity.

• Interfacing with Arduino Uno (Analog Input A0):

- The output from the op-amp is fed into the **analog pin A0** of the Arduino Uno.
- Arduino continuously reads the incoming signal and forwards it to a computer or peripheral system via **serial communication**.
- The microcontroller acts as a real-time interface between the raw EEG signal and higher-level applications.

2. Signal Processing & Classification

This step focuses on interpreting the raw EEG signals and converting them into meaningful data for action triggers (like blink detection or concentration monitoring).

• Capturing EEG Signals and Filtering Noise:

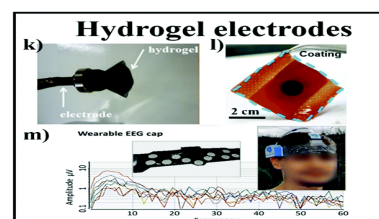
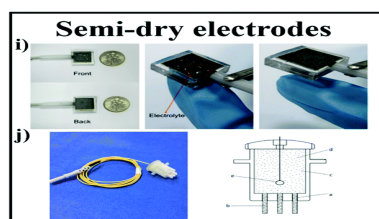
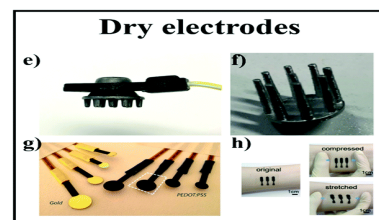
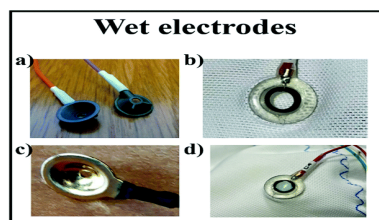
- The EEG signal is sampled by the Arduino and sent to a computer.
- Since EEG is easily contaminated by environmental noise (e.g., powerline interference), **basic filtering** is applied.
- Filtering can be done through:
 - **Hardware filters** (in the op-amp circuit),
 - Or **software filtering** (using moving average or digital low-pass filters).

• Detecting Eye Blinks using Time-Domain Analysis:

- Eye blinks produce **sharp, high-amplitude spikes** in EEG signals, especially on the frontal lobe.
- In the code, a **threshold value** (e.g., 550) is used to detect these spikes.
- If the EEG value exceeds the threshold, and the time between peaks is appropriate (debouncing), it's registered as a **blink**.
- Multiple blinks in a time window (like 2 blinks within 1.5 seconds) are interpreted as control signals (e.g., turn on light or music).

• Identifying Concentration/Relaxation Based on Frequency/Power Analysis:

- While the current system uses **simple thresholding**, advanced BCI systems analyze brainwave frequency bands:
 - **Alpha waves (8–13 Hz)** → relaxation
 - **Beta waves (13–30 Hz)** → focus/concentration
 - **Theta/Delta** → drowsiness or sleep
- In this project, thresholds like **>600** indicate **high focus**, and **<400** suggest **relaxation**.
- These thresholds are calibrated based on empirical testing and used to trigger contextual actions like playing white noise or turning on a fan.



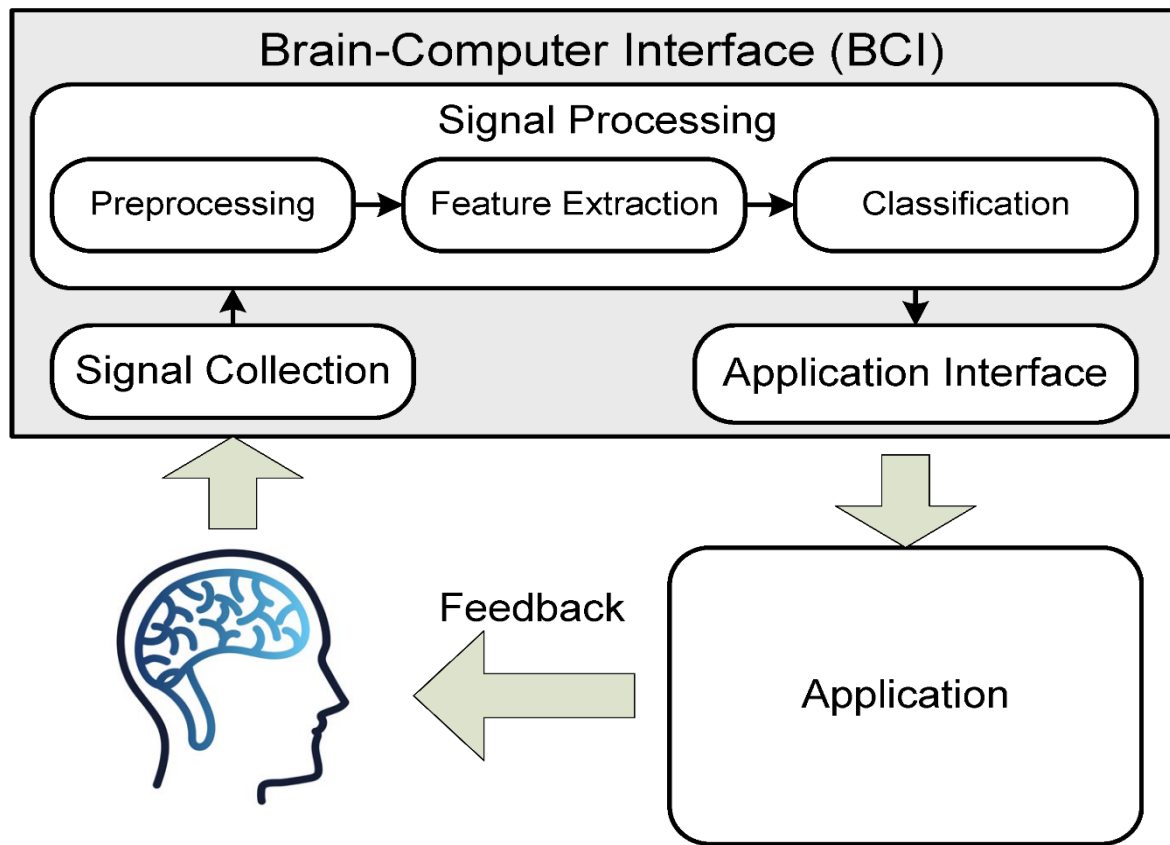


Fig 1.4 Flow Diagram of signals

3. Software & Integration

This phase is where the hardware capabilities of the system are tied to intelligent, responsive software to create a seamless Brain-Computer Interface experience.

• Writing Arduino Code to Analyze Signals in Real-Time:

- The Arduino Uno continuously reads analog values from the EEG electrodes connected via the op-amp.
- Using real-time logic, the code detects spikes (eye blinks), monitors signal patterns, and differentiates between multiple blinks and signal intensities.
- It keeps track of blink counts, timing windows, and threshold crossings using built-in functions like `millis()` to measure time differences.
- It then sends interpreted data (e.g., "BLINK2", "MULTIBLINK", or numeric EEG values) to the Python interface through serial communication for further action.

• Building a Python-Based AI Assistant ("Rocker") Capable Of:

- **Interpreting Brain States:**
 - The Python script listens to serial data from the Arduino.
 - When a pattern like "BLINK2" or EEG values crossing certain thresholds is detected, it understands the user's intent or mental state (e.g., relaxed, focused).
- **Executing Commands:**
 - Depending on detected brain states, Rocker sends back commands like `FAN_ON` or `LED_OFF` to the Arduino to control physical devices.
 - It also handles multimedia control like playing or stopping music, or initiating focus modes using white noise.
- **Providing Verbal Feedback or Suggestions:**

- Using pytsx3, Rocker speaks to the user to acknowledge actions or offer suggestions.
- For example: "You look relaxed. Starting the fan," or "You are focused. Enabling focus mode."
- This conversational feedback loop creates a more interactive and human-like experience.

4. Testing & Demonstration

This critical phase ensures the practical viability of the system and showcases its real-world application.

• Real-Time Control of Appliances and Systems Based on Mental State:

- The system was tested in real-time conditions to verify whether blinks and mental states reliably trigger the desired actions.
- For instance, two blinks within 1.5 seconds would turn on the light, while focused states would activate white noise.
- This demonstrates the feasibility of using brain signals alone to control common devices without any touch or speech input.

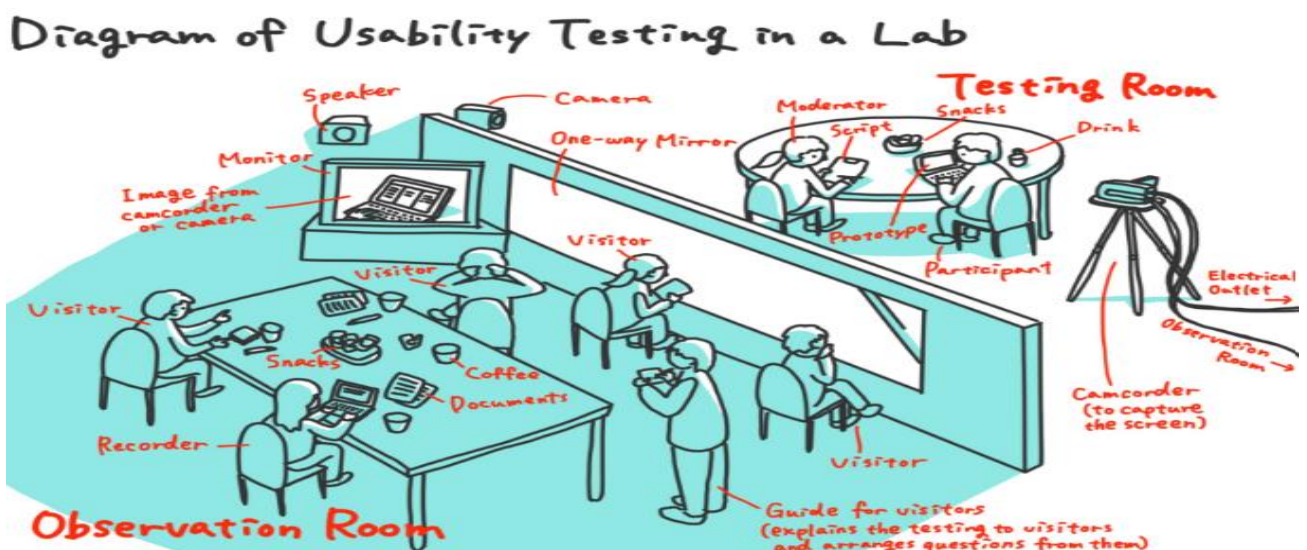
• Logging Responses for Evaluation:

- All incoming EEG values and interpreted events (like blink detection or command execution) are logged in the terminal or stored in files for post-session analysis.
- This helps in fine-tuning thresholds, verifying system behavior, and analyzing user response patterns to improve performance and accuracy.

• Conducting Usability Tests in Home/Lab Environments:

- Multiple usability sessions were carried out in both controlled (lab) and uncontrolled (home) environments.
- These tests were essential to assess:
 - Signal consistency across different users,
 - Response time and system reliability,
 - Comfort and intuitiveness of using the system.
- Feedback from these tests helped guide improvements in hardware (like better electrode placement) and software (like refining logic and feedback tone). This project avoids invasive methods, limits EEG interpretation to basic cognitive states, and focuses on **cost-effective, safe, and impactful innovation**.

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2. Literature Review

2.1 Existing BCI Technologies

Brain-Computer Interfaces (BCIs) represent a revolutionary technology that bridges the neural activity of the human brain and external devices, allowing direct communication without muscular interaction. Over the past few decades, various BCI paradigms have emerged, primarily classified as **invasive**, **semi-invasive**, and **non-invasive** systems. Invasive BCIs involve direct implantation of electrodes into the cerebral cortex (e.g., Utah Array), offering high-resolution data but posing ethical and medical risks. Semi-invasive methods position electrodes inside the skull but outside the brain tissue, while non-invasive BCIs—most notably **Electroencephalography (EEG)** based systems—detect signals from the scalp surface and are most widely used for consumer and research applications.

Among the commercially available non-invasive solutions, **Emotiv Epoc**, **NeuroSky MindWave**, and **OpenBCI** are noteworthy. Emotiv offers 14-channel EEG headsets suitable for research and application prototyping, while NeuroSky provides a single-channel consumer-grade device. OpenBCI stands out for its open hardware philosophy, modular designs, and community-driven ecosystem. These devices support a broad range of applications:

- **Medical Assistive Technology:** Enabling locked-in patients (e.g., ALS) to communicate or control wheelchairs.
- **Neurorehabilitation:** Assisting stroke patients with motor imagery feedback.
- **Entertainment:** Creating immersive experiences in gaming, VR, and media through brain-driven inputs.

Despite these advancements, challenges persist in signal fidelity, high latency, limited control bandwidth, and user fatigue. Most devices require expert setup and signal processing expertise, limiting scalability and daily usability.

2.2 Open-Source EEG Systems

The democratization of BCI research owes much to open-source platforms, particularly **OpenBCI**, **NeuroPyte**, and **BrainFlow**. These platforms provide both hardware and software tools that enable cost-effective experimentation and application development. For instance:

- **OpenBCI's Cyton Board** offers 8–16 channels of EEG acquisition and is compatible with various electrode types (wet, dry, or saline).
- **NeuroPyte** is a flexible real-time signal processing suite built for neurotechnological applications, often paired with machine learning modules.
- **BrainFlow** provides APIs for streaming, filtering, and visualizing biosignals in multiple programming languages (Python, JavaScript, C++).

These tools have made it possible to build highly creative BCI projects like **brain-controlled drones**, **mind-operated musical instruments**, and **mental workload trackers**. However, they still suffer from the following issues:

- **Hardware Complexity:** Requires assembling headsets and soldering circuits, which can be daunting for beginners.
- **Signal Quality Variance:** Particularly in dry electrodes, due to poor skin contact and environmental noise.
- **Scalability:** Real-time applications on low-resource microcontrollers (like Arduino Uno) face limitations in computation and memory.

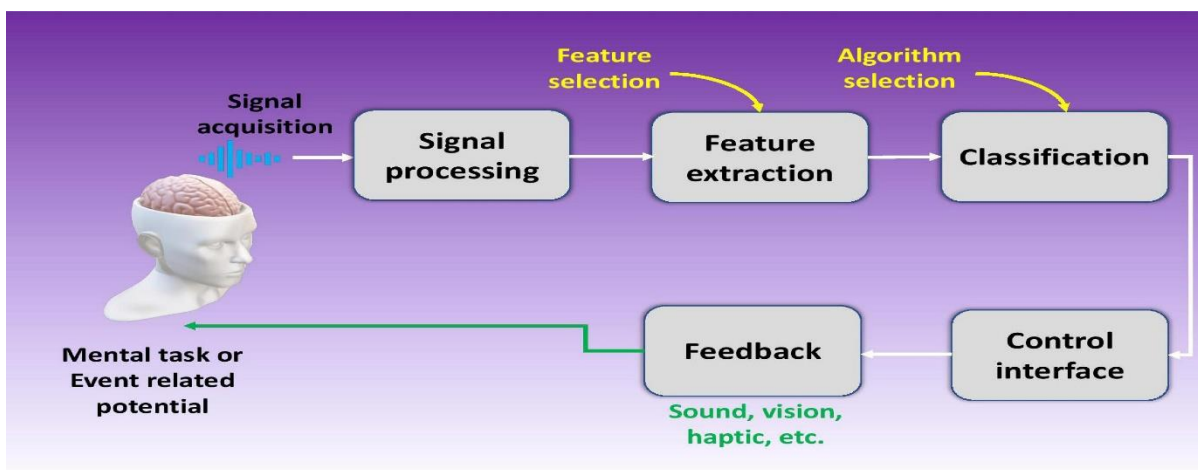


Fig 2.1 EEG setup

Our project aims to bridge this gap by building a minimal, affordable, and customizable EEG interface that retains meaningful signal resolution while enabling real-time interaction using AI tools like “Rocker”—our embedded assistant.

2.3 Signal Acquisition Techniques

The core of any Brain-Computer Interface (BCI) lies in accurate and reliable acquisition of neural signals, particularly Electroencephalogram (EEG) data. EEG captures the summed electrical activity of post-synaptic potentials from pyramidal neurons in the cortex, measured non-invasively using surface electrodes placed on the scalp.



Electrode Placement: The 10–20 International System

Electrode locations are standardized using the **International 10–20 system**, ensuring reproducibility and targeting of specific brain regions. Common positions include:

- **Fp1, Fp2** (frontal poles): Linked with eye blinks, attention, and executive function.
- **Cz** (central vertex): Associated with motor control.
- **O1, O2** (occipital): Visual processing.
- **T3/T4** (temporal): Auditory and emotional response.

Our prototype leverages **Fp1 and Fp2**, ideal for capturing **attention and voluntary blink patterns**, crucial for simplified control signals in BCI applications.

Electrode Types

Type	Description	Use Case
Wet Electrodes	Use conductive gel for low impedance (~5–10 kΩ), high-quality signal	Clinical-grade EEG, research
Dry Electrodes	Conductive rubber/metal, no gel; higher impedance (~10–50 kΩ)	DIY, wearable EEG setups
Textile/Printed	Flexible, washable electrodes integrated into fabrics	Next-gen wearables, neurotech
Homemade Coins	Nickel-coated coins used with elastic bands and shielding (our innovation)	Ultra-low-cost BCI prototyping

T.2.1 Electrode Types

Our design uses **nickel coins** and **jumper wires** as dry electrodes, forming a functional yet cost-effective setup ideal for experimentation and educational purposes.

EEG Signal Characteristics

EEG captures neural oscillations across a range of frequency bands, each associated with distinct cognitive and physiological states:

Frequency Band	Range	Cognitive State
Delta	0.5 – 4 Hz	Deep sleep, unconscious states
Theta	4 – 8 Hz	Drowsiness, meditation
Alpha	8 – 12 Hz	Relaxation, passive focus
Beta	12 – 30 Hz	Active focus, problem-solving
Gamma	>30 Hz	Complex processing, perception

T.2.2 Frequency Types

Understanding these bands allows tailoring of filters and feature extractors to specific applications (e.g., relaxation apps, attention monitoring, seizure detection).

Signal Conditioning Pipeline

Because EEG signals are extremely low amplitude (10–100 μV) and prone to noise, the signal acquisition pipeline must be carefully engineered:

1. Amplification

EEG signals are amplified 1000x or more using:

- **Instrumentation Amplifiers** (e.g., **AD620**, **AD623**) for differential gain and high CMRR.
- **Op-Amps** (e.g., LM358, TL072) for custom gain stages.
This ensures signal fidelity without saturation or clipping.

2. Filtering

A **bandpass filter (0.5–40 Hz)** is used to isolate relevant brain wave frequencies, while removing:

- **Low-frequency drift** (motion artifacts, sweat).
- **High-frequency noise** (EMG, electronics).
Optionally, **Notch filters (50/60 Hz)** remove power line interference.

3. Artifact Removal

EOG (Eye blink) and EMG (muscle twitch) artifacts are common in frontal recordings. Common techniques to mitigate these include:

- **Adaptive filtering** or
- **Independent Component Analysis (ICA)** in post-processing.
Our DIY system uses signal modeling combined with dynamic thresholding to detect and utilize blink artifacts as a feature, not noise.

4. Digitization

The filtered analog EEG is digitized using:

- **Arduino Uno (10-bit ADC @ ~256 Hz)**: Basic, affordable solution.
 - **ESP32 (12-bit ADC)**: Higher resolution and speed, suitable for more advanced setups.
Sampling rate is selected to match the Nyquist criterion for EEG (typically ≥ 128 Hz).
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DIY Implementation Strategy (Our Approach)

We employ **homemade nickel-coated coins** as electrodes, which are connected using jumper wires to an **Op-Amp** circuit for signal amplification. The signals are then fed into the **analog pin (A0)** of an **Arduino Uno**. The simplicity of this setup is complemented by its affordability, making it an ideal solution for DIY EEG-based applications.

By following this approach, we are able to create a **low-cost and accessible EEG acquisition system** that maintains signal fidelity and provides the necessary features for practical brain-computer interface experiments.

Our system uses **coin electrodes** interfaced through Op-Amps and connected to Arduino's analog pin (A0), combining affordability with accessibility. Signal processing is done in Python or Node.js to extract features like blink rate or concentration level.

2.4 Research Gaps Identified

Although **Brain-Computer Interface (BCI)** research has made significant strides in recent years, there are still numerous areas that remain underexplored, especially when it comes to creating **affordable, portable, and AI-integrated solutions**. While traditional high-resolution systems provide great accuracy, they are **expensive** and out of reach for most individual users, hobbyists, and educational setups. On the other hand, lower-cost systems often suffer from **reduced performance** and **limited functionalities**. Addressing these gaps is essential for making BCIs widely accessible and practically useful across various domains, from assistive technology to personal automation. Below are some key **research gaps** that our project aims to address:

1. Cost vs. Performance Trade-off

- **High-Resolution vs. Affordable Systems**
High-resolution EEG systems, which offer **multiple channels** and **accurate brainwave monitoring**, can cost over \$500 or more. These systems are generally used in research or clinical settings, where accuracy is a priority. However, for everyday users, students, or hobbyists, the cost of such systems is prohibitive. On the other hand, affordable consumer-grade EEG systems provide only a **few channels** (e.g., 1–4 channels) and often deliver **low accuracy**, making them less reliable for precise mental state detection.
- **The Challenge:**
A major gap in BCI research is finding a way to balance **cost** and **performance**. How can a system be made affordable (below ₹1000) while still providing sufficient **signal fidelity** and **functional versatility**? For example, a system with 8–16 channels might provide more reliable data than a system with only a single channel, but it should still remain accessible. Our project addresses this challenge by focusing on **low-cost electrodes**, **open-source hardware**, and **simplified signal processing techniques** that can deliver adequate performance for basic mental state detection.



2. Real-Time Processing on Edge Devices

- **Current Dependence on PCs for Processing**

Most commercial BCI systems require a connection to a **PC** or **external computing device** for processing EEG data. While this setup allows for powerful processing capabilities, it reduces the system's **portability** and **real-time responsiveness**. In many cases, the **latency** introduced by transmitting data to a remote device can hinder the **real-time nature** of BCI applications, particularly those that require immediate feedback for controlling devices or interacting with the environment.

- **The Challenge:**

Achieving **real-time signal processing** directly on **edge devices** like **Arduino**, **ESP32**, or **Raspberry Pi** is a significant technical hurdle. These devices have **limited processing power** and **memory capacity**, making it difficult to handle the computational demands of **EEG signal interpretation**, **machine learning models**, and **control logic**—all with minimal **latency** and **low power consumption**. Our project addresses this gap by implementing **simple yet effective algorithms** that can run efficiently on microcontrollers, allowing for **low-latency signal processing** and **control outputs** without the need for external computers.

3. Noise Robustness in DIY Systems

- **Challenges in DIY BCI Setups**

Many hobbyist-grade BCI systems are built using **off-the-shelf** components or **homemade setups**, which often lack the sophisticated **shielding**, **grounding**, and **signal filtering** mechanisms required to minimize noise interference. Common issues include **electromagnetic interference (EMI)** from nearby electronic devices, **motion artifacts** from muscle activity, and **poor electrode contact** with the scalp, all of which can lead to **noisy** or **unreliable EEG signals**.

- **The Challenge:**

Ensuring **signal fidelity** in low-cost, DIY EEG systems is an ongoing challenge. Without proper **noise mitigation strategies**, the EEG signals may become corrupted, making it impossible to reliably detect mental states like **focus**, **relaxation**, or **blinks**. Most existing hobbyist systems do not incorporate effective techniques for minimizing these issues, leading to **inconsistent** or **inaccurate** results. In this project, we propose integrating **shielded electrodes**, **RC snubber networks**, and **adaptive filtering algorithms** to enhance **noise robustness**, allowing for more stable and reliable EEG readings in a DIY context.

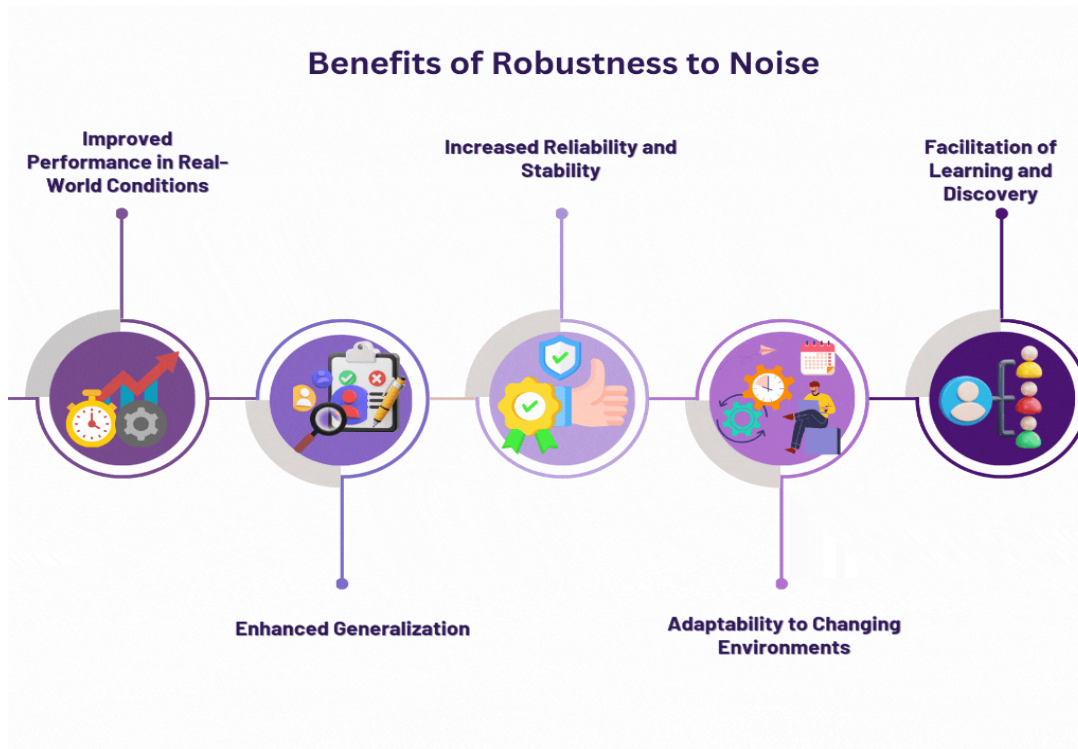


Fig 2.2 Benefit to Noise Robustness

4. AI-Based Multimodal Fusion

- **Limitations of Current EEG-Based BCIs**

Existing BCI systems predominantly rely on **EEG data alone** to interpret brain states. While this is effective for basic applications, it is often limited in its ability to recognize complex **mental states** or integrate with external systems (e.g., smart devices, voice assistants). Moreover, these systems lack integration with other **modalities**, such as **speech recognition** or **context-aware computing**, which could significantly enhance their functionality.

- **The Challenge:**

There is a growing need for **multimodal fusion** in BCIs, where data from different sources (e.g., EEG, speech, motion sensors) is combined to create a more robust and context-aware system. For example, integrating **voice commands** with **mental state detection** could lead to more **adaptive** and **personalized** user experiences. Our project specifically addresses this gap by creating a hybrid system that integrates **EEG signals** with a **Natural Language Processing (NLP) assistant** named "**Rocker**." This system not only responds to **mental states** but also **voice inputs**, enabling a truly **multimodal** interaction.

5. Lack of Emotion and Cognitive State Personalization

- **Generic Thresholds and One-Size-Fits-All Models**

Many current BCI systems rely on **static thresholds** for detecting states like **focus**, **relaxation**, and **stress**, which are often calibrated based on **generalized** or **average population data**. However, the way individuals experience these states can vary widely due to differences in **brain anatomy**, **cognitive processing**, and **external influences**. Personalizing these thresholds for each user is critical for creating a system that accurately interprets and responds to their unique mental states.

- **The Challenge:**

There is a lack of **personalized models** that adapt to the **individual's unique cognitive and emotional patterns**. Most systems offer fixed thresholds for detecting **mental states**, but these may not be accurate for all users. **Machine learning**

can play a key role in this, allowing the system to **learn** the user's specific brainwave patterns and adjust the thresholds accordingly. Our project seeks to explore this by incorporating **adaptive machine learning models** that **personalize** the detection of **relaxation**, **focus**, and other cognitive states, making the BCI more **accurate** and **responsive** for different users.

Vision for the Future of BCI

By addressing the research gaps mentioned above, our project aims to develop a **hybrid BCI system** that is:

- **Modular:** Both the hardware and software components are designed to be easily **upgradable**, allowing users to adapt and improve the system over time.
- **Intelligent:** By incorporating **adaptive AI** and **machine learning**, the system can **personalize** its responses and **optimize** performance based on the user's unique mental states.
- **Affordable:** We aim to keep the total **build cost under ₹1000**, making this system accessible to a wide range of users, including hobbyists, students, and researchers.
- **Contextually Aware:** The system will not only interpret brain signals but also integrate with **voice inputs** and other **sensor data** to provide a more **interactive and dynamic** user experience.

In conclusion, this project aims to bridge the gap between **cost-effective** and **high-performance** BCIs, while also pushing the boundaries of **AI integration** and **personalized cognitive state detection**. Through this, we hope to make BCIs more **accessible**, **adaptable**, and **contextually aware** for a broader audience, opening up new possibilities for **assistive technologies**, **brain-controlled automation**, and **human-computer interaction**.

